Divya Chari

Physics PhD Candidate, The Ohio State University

chari.16@osu.edu | divyachari01.github.io | github.com/divyachari01 | linkedin.com/in/divya-chari-13325914a

EDUCATION

Ph.D. in Physics, The Ohio State University 2023 to present

Advisors: Prof. Anil K. Pradhan and Prof. Sultana N. Nahar. Candidacy exam completed.

M.S. in Physics, The Ohio State University 2026

B.Sc. Honours, Physics and Astronomy, University of British Columbia 2018 to 2023

RESEARCH EXPERIENCE

Graduate Research Assistant, The Ohio State University 2023 to present

R-Matrix Opacity Project (RMOP), Department of Astronomy and CCAPP

- Compute photoionization cross sections and opacities with the Breit-Pauli R-matrix (BPRM) method, for the solar opacity problem, helioseismology and high-energy-density plasma experiments.
- Model plasma broadening of autoionizing resonances (electron collisions, Stark, Doppler and free-free effects) as a function of temperature and electron density, and carry it through to monochromatic and Rosseland mean opacities for stellar mixtures.
- Compare equation-of-state results with opacity calculations, in collaboration with outside researchers.
- Develop and maintain Fortran, Python and C++ pipelines for large-scale opacity calculations on high-performance computing clusters (Ohio Supercomputer Center, OSU Unity).

SOFTWARE

ssopt: automated atomic structure calculations with the SUPERSTRUCTURE code 2026

github.com/divyachari01/ssopt | Python, Fortran, Slurm | with S. N. Nahar and A. K. Pradhan | CC BY-NC 4.0

- Given only an ion's name, the code builds a configuration list that is larger than NIST's: it starts from every NIST configuration and adds single and paired double excitations from the ground configuration, up to n = 4 and l = 3. It then tunes the scaling parameters of SUPERSTRUCTURE against NIST energy levels, with no settings specific to any ion.
- A Gaussian process surrogate model keeps this to about 300 structure runs per ion, using two-stage tuning (per angular momentum, then per orbital). The same settings worked for 96 ions, with a median energy error of about 1% against NIST.
- The code then checks transition rates against NIST and NORAD and against measured lifetimes, relabels levels to match NIST, and exports level and transition tables in NORAD format.

PUBLICATIONS

Chari, D., Nahar, S. N. and Pradhan, A. K. R-matrix calculations for opacities: V. Temperature-density dependence of photoabsorption cross sections and opacity spectra of oxygen ions O VI and O VII. *J. Phys. B: At. Mol. Opt. Phys.* 59, 145003 (2026). doi:10.1088/1361-6455/ae8c98

PRESENTATIONS

17th Fundamental Science with Pulsed Power Workshop (ZFS), Sandia National Laboratories (poster) 2026

APS Division of Atomic, Molecular and Optical Physics (DAMOP): *Temperature-density effects on atomic cross sections in high-energy-density plasmas* 2026

Ohio Supercomputer Center (OSC) Research Symposium, short talk 2026

APS Division of Atomic, Molecular and Optical Physics (DAMOP): *The Opacity Project: R-matrix calculations, plasma broadening effects and equation-of-state for high-energy-density sources* 2025

TEACHING EXPERIENCE

Teaching Assistant and Grader, The Ohio State University 2023 to present

PHYS 1200/1201 (introductory), 5500/5501 (undergraduate quantum mechanics), 7502 (graduate quantum mechanics); duties vary by semester

- Teach undergraduate physics labs and recitations, give feedback on lab reports, and guide students through quantitative reasoning and data interpretation.
- Grade coursework in Gradescope and develop rubrics for consistent, fair assessment.

Teaching Assistant, University of British Columbia (physics labs and recitations) 2021 to 2023